Computational Fluid Dynamics - 5/7/2023

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Exercise 1.

Consider the following Navier-Stokes problem on the square $\Omega = [0,1]^2$:

$$\begin{cases} \frac{\partial \mathbf{u}}{\partial t} - \nu \triangle \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{u} + \nabla p = \mathbf{f} & \text{in} \quad \Omega, \quad t > 0 \\ \operatorname{div} \mathbf{u} = \mathbf{0} & \text{in} \quad \Omega, \quad t > 0, \\ \mathbf{u} = \mathbf{u}_0 & \text{in} \quad \Omega, \quad t = 0, \\ -p\mathbf{n} + \nu \nabla \mathbf{u} \cdot \mathbf{n} = \mathbf{h} & \text{on} \quad \Gamma_N = \{\mathbf{x} = (x,y) \in \partial\Omega : y = 0\}, \quad t > 0, \\ \mathbf{u} = \mathbf{g} & \text{on} \quad \Gamma_D = \partial\Omega \setminus \Gamma_N, \quad t > 0, \end{cases} \tag{1}$$

where $\nu = 0.01$ and the forcing term $\mathbf{f}$ is given by

$$\mathbf{f} = \begin{pmatrix} -\cos(x)\sin(y)(\cos(t) + 2\nu\sin(t)) \\ \sin(x)\cos(y)(\cos(t) + 2\nu\sin(t)) \end{pmatrix}.$$

Problem (1) has the following exact solution

$$\mathbf{u}_{\text{ex}} = \begin{pmatrix} -\cos(x)\sin(y)\sin(t) \\ \sin(x)\cos(y)\sin(t) \end{pmatrix}, \qquad p_{\text{ex}} = -\frac{1}{4}(\cos(2x) + \cos(2y))(\sin(t))^2,$$

with the Dirichlet datum $\mathbf{g}$ and the Neumann datum $\mathbf{h}$ that can be computed based on the exact solution.

a) Write the semi-discrete nonlinear problem obtained from a Crank-Nicolson time discretization of Problem (1), with an implicit treatment of the convection term.

b) Introduce a suitable linearization of the semi-discrete problem based on the Newton method to solve the nonlinear problem corresponding to each time iteration. Employ a suitable stop criterion based on the increment.

c) Write the weak form of the (linear) problem corresponding to each Newton iteration.

d) Write the fully discrete problem obtained by a finite element discretization in space, with $\mathbb{P}_2 - \mathbb{P}_1$ finite element spaces.

e) Write a FreeFEM++ solver, starting from the template `exam.edp` for the solution of Problem (1).

f) Using the solver, assess the convergence and stability properties of the proposed scheme. In particular, solve the problem in the time interval $t \in (0,1]$ on a uniform grid with $h = 0.1$ and with different time steps $\Delta t = \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}$. Fill the following table with the pressure error in the $L^\infty(0,1;L^2(\Omega))$ norm and with the velocity error in the $L^\infty(0,1;[H^1(\Omega)]^2)$ norm.

Δt	$\Vert \mathrm{u} - \mathrm{u}_{\text{ex}} \Vert_{L^\infty(0,1;[H^1(\Omega)]^2)}$	$\Vert p - p_{\text{ex}} \Vert_{L^\infty(0,1;L^2(\Omega))}$
$\frac{1}{2}$		
$\frac{1}{4}$		
$\frac{1}{8}$		
$\frac{1}{16}$		

g) Estimate numerically the convergence rates and discuss the results with respect to the theoretical estimates.